

Лабораторная работа № 2. Измерение и тестирование пропускной способности сети.Интерактивный эксперимент

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Цель работы

Основной целью работы является знакомство с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получение навыков проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.

Задание

1. Установить на виртуальную машину mininet iPerf3 и дополнительное программное обеспечения для визуализации и обработки данных.
2. Провести ряд интерактивных экспериментов по измерению пропускной способности с помощью iPerf3 с построением графиков.

Выполнение лабораторной работы

Установка необходимого программного обеспечения

```
ubuntu@20.04.1 LTS:~$ mininet-um ttyl
mininet-um login: mininet
Password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

Last login: Wed Sep 10 07:09:13 PDT 2025 from 192.168.56.1 on pts/1
mininet@mininet-um:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
    ether 08:00:27:40:a4:3f txqueuelen 1000 (Ethernet)
    RX packets 3 bytes 1240 (1.2 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2 bytes 684 (684.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

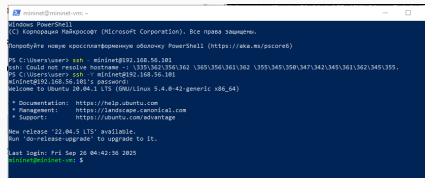
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    ether 08:00:27:19:7d:b7 txqueuelen 1000 (Ethernet)
    RX packets 107 bytes 21444 (21.4 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 105 bytes 10309 (10.3 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 28 bytes 2912 (2.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 2912 (2.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet@mininet-um:~$ _
```

Рис. 1: Виртуальная машина запущена

Установка необходимого программного обеспечения



```
mininet@mininet-vm: -
Windows PowerShell
(C) Корпорация Майкрософт (Microsoft Corporation). Все права защищены.
Получите новую кроссплатформенную оболочку PowerShell (https://aka.ms/powershell)

PS C:\Users\user> ssh -l mininet 192.168.56.101
ssh: Could not resolve hostname -l: 1335:362:356:362: 1365:354:363:362: 1355:345:350:347:342:345:363:362:345:355.
PS C:\Users\user> ssh -V mininet@192.168.56.101
mininet@192.168.56.101's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-42-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

New release '22.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Fri Sep 26 04:42:36 2025
mininet@mininet-vm: $
```

Рис. 2: Подключение к виртуальной машине с основной ОС

Установка необходимого программного обеспечения

```
mininet@mininet-vm: $ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.56.101 netmask 255.255.255.0 broadcast 192.168.56.255
    ether 08:00:27:40:a4:3f txqueuelen 1000 (Ethernet)
    RX packets 49 bytes 8598 (8.5 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 42 bytes 7477 (7.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

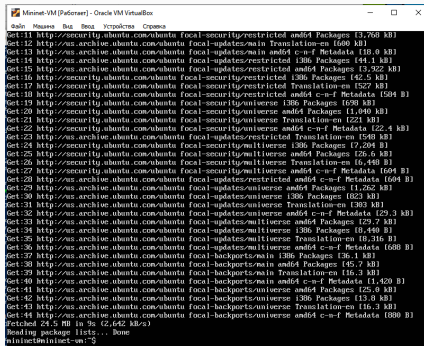
eth1: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.0.2.15 netmask 255.255.255.0 broadcast 10.0.2.255
    ether 08:00:27:19:7d:07 txqueuelen 1000 (Ethernet)
    RX packets 289 bytes 37044 (37.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 304 bytes 27519 (27.5 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    loop txqueuelen 1000 (Local Loopback)
    RX packets 28 bytes 2912 (2.9 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 28 bytes 2912 (2.9 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

mininet@mininet-vm: $
```

Рис. 3: IP-адреса виртуальной машины

Установка необходимого программного обеспечения



```
Mininet-VM (Pafnarr) - Oracle VM VirtualBox
ash@Mininet: ~$ dpkg --get-urls
Get:11 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 Packages [3,768 kB]
Get:12 http://us.archive.ubuntu.com/ubuntu focal-updates/main Translation-en [690 kB]
Get:13 http://us.archive.ubuntu.com/ubuntu focal-updates/main amd64 c-n-f Metadata [18.0 kB]
Get:14 http://us.archive.ubuntu.com/ubuntu focal-updates/restricted 1386 Packages [44.1 kB]
Get:15 http://us.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 Packages [13,322 kB]
Get:16 http://security.ubuntu.com/ubuntu focal-security/restricted 1386 Packages [42.5 kB]
Get:17 http://security.ubuntu.com/ubuntu focal-security/restricted Translation-en [527 kB]
Get:18 http://security.ubuntu.com/ubuntu focal-security/restricted amd64 c-n-f Metadata [504 B]
Get:19 http://security.ubuntu.com/ubuntu focal-security/universe 1386 Packages [693 kB]
Get:20 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages [11,040 kB]
Get:21 http://security.ubuntu.com/ubuntu focal-security/universe Translation-en [221 kB]
Get:22 http://security.ubuntu.com/ubuntu focal-security/universe amd64 c-n-f Metadata [22.4 kB]
Get:23 http://us.archive.ubuntu.com/ubuntu focal-updates/restricted Translation-en [548 kB]
Get:24 http://security.ubuntu.com/ubuntu focal-security/multiverse 1386 Packages [7,204 B]
Get:25 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 Packages [26.6 kB]
Get:26 http://security.ubuntu.com/ubuntu focal-security/multiverse Translation-en [6,448 B]
Get:27 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64 c-n-f Metadata [604 B]
Get:28 http://us.archive.ubuntu.com/ubuntu focal-updates/restricted amd64 c-n-f Metadata [604 B]
Get:29 http://us.archive.ubuntu.com/ubuntu focal-updates/universe amd64 Packages [11,262 kB]
Get:30 http://us.archive.ubuntu.com/ubuntu focal-updates/universe 1386 Packages [823 kB]
Get:31 http://us.archive.ubuntu.com/ubuntu focal-updates/universe Translation-en [393 kB]
Get:32 http://us.archive.ubuntu.com/ubuntu focal-updates/universe amd64 c-n-f Metadata [29.3 kB]
Get:33 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 Packages [29.7 kB]
Get:34 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse 1386 Packages [8,440 B]
Get:35 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse Translation-en [9,516 B]
Get:36 http://us.archive.ubuntu.com/ubuntu focal-updates/multiverse amd64 c-n-f Metadata [688 B]
Get:37 http://us.archive.ubuntu.com/ubuntu focal-backports/main 1386 Packages [136.1 kB]
Get:38 http://us.archive.ubuntu.com/ubuntu focal-backports/main amd64 Packages [45.7 kB]
Get:39 http://us.archive.ubuntu.com/ubuntu focal-backports/main Translation-en [116.3 kB]
Get:40 http://us.archive.ubuntu.com/ubuntu focal-backports/main amd64 c-n-f Metadata [11,420 B]
Get:41 http://us.archive.ubuntu.com/ubuntu focal-backports/universe amd64 Packages [125.0 kB]
Get:42 http://us.archive.ubuntu.com/ubuntu focal-backports/universe 1386 Packages [13.8 kB]
Get:43 http://us.archive.ubuntu.com/ubuntu focal-backports/universe Translation-en [116.3 kB]
Get:44 http://us.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata [880 B]
Fetched 24.5 MB in 3s (7,642 kB/s)
Reading package lists... Done
mininet@mininet-uni:~$
```

Рис. 4: Обновление репозитория ПО

Установка необходимого программного обеспечения

```
Get:44 http://us.archive.ubuntu.com/ubuntu focal-backports/universe amd64 c-n-f Metadata (100 B)
Fetched 24.5 MB in 9s (2,642 kB/s)
Reading package lists... Done
mininet@mininet-vm:~$ sudo apt-get install iperf3
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  libiperf0 libhsctp
Suggested packages:
  libhsctp-tools
The following NEW packages will be installed:
  iperf3 libiperf0 libhsctp
0 upgraded, 3 newly installed, 0 to remove and 395 not upgraded.
Need to get 94.1 kB of archives:
after this operation, 331 kB of additional disk space will be used.
Do you want to continue? (Y/n) y
Get:1 http://us.archive.ubuntu.com/ubuntu focal/main amd64 libhsctp1 amd64 1.0.10-dfsg-1 [7,876 B]
Get:2 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 libiperf0 amd64 3.7-3 [72.0 kB]
Get:3 http://us.archive.ubuntu.com/ubuntu focal/universe amd64 iperf3 amd64 3.7-3 [14.2 kB]
Fetched 94.1 kB in 1s (114 kB/s)
Selecting previously unselected package libhsctp1:amd64.
(Reading database ... 102146 files and directories currently installed.)
Preparing to unpack .../libhsctp1_1.0.10-dfsg-1_amd64.deb ...
Unpacking libhsctp1:amd64 (1.0.10-dfsg-1) ...
Selecting previously unselected package libiperf0:amd64.
Preparing to unpack .../libiperf0_3.7-3_amd64.deb ...
Unpacking libiperf0:amd64 (3.7-3) ...
Selecting previously unselected package iperf3.
Preparing to unpack .../iperf3_3.7-3_amd64.deb ...
Unpacking iperf3 (3.7-3) ...
Setting up libhsctp1:amd64 (1.0.10-dfsg-1) ...
Setting up libiperf0:amd64 (3.7-3) ...
Setting up iperf3 (3.7-3) ...
Processing triggers for man-db (2.9.1-1) ...
Processing triggers for libc-bin (2.31-0ubuntu9) ...
mininet@mininet-vm:~$
```

Активация Windows

Рис. 5: Установка iperf3

Установка необходимого программного обеспечения

```
minic@minic:~$ sudo apt-get install git jq gnuplot-nox evince
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following additional packages will be installed:
  gdlra aspell aspell-en bubblewrap enchant-2 evince-common fonts-liberation gnome-desktop3-data
  gnuplot-data groff hunspell-en-us imagemagick imagemagick-6.q16 libarchive13 libaspell15
  libdjvulibre-text libdjvulibre21 libenchant-2-2 libeddocument3-4 libevview3-3
  libgnome-desktop-3-19 libgspell1-1-2 libgspell1-common libgxps2 libhunspell-1.7-0 libliblev3
  libjq1 libkpathsea6 liblua5.3-0 libmagickcore-6.q16-6-extra libnautilus-extension1a libnetpbm10
  libopenjre liboss3 libpng5 libpopt0 libpopt24 libpoppler-glib0 libpoppler97 libsecret-1-0
  libsecret-common libspectre1 libsynctex2 libunf0.2-7 netpbm putils
Suggested packages:
  aspell-doc spellutils gnu's nautilus-sendto unrar git-daemon-run | git-daemon-sysvinit git-doc
  git-el git-email git-gui gitk git-cvs git-mediawiki git-svn gnuplot-doc hunspell
  openoffice.org-hunspell | openoffice.org-core imagemagick-doc antitrac cups-bd | lpr | lprng
  curl encrypt ffmpeg gimp grads graphviz hp2xx htm2ps libaaf-bin mlayer povray radianse
  sane-utils texlive-base-bin transfig ufwu-batch xdg-utils lrzip libenchant-2-voikko linkscape
  libjxr-tools libunf0.2-7-gtk
The following NEW packages will be installed:
  gdlra aspell aspell-en bubblewrap enchant-2 evince evince-common fonts-liberation
  gnome-desktop3-data gnuplot-data gnuplot-nox groff hunspell-en-us imagemagick imagemagick-6.q16
  jq libarchive13 libaspell15 libdjvulibre-text libdjvulibre21 libenchant-2-2 libeddocument3-4
  libevview3-3 libgnome-desktop-3-19 libgspell1-1-2 libgspell1-common libgxps2 libhunspell-1.7-0
  libliblev3 libjq1 libkpathsea6 liblua5.3-0 libmagickcore-6.q16-6-extra libnautilus-extension1a
  libnetpbm10 libopenjre liboss3 libpng5 libpopt0 libpopt24 libpoppler-glib0 libpoppler97 libsecret-1-0
  libsecret-common libspectre1 libsynctex2 libunf0.2-7 netpbm putils
The following packages will be upgraded:
  git
1 upgraded, 48 newly installed, 0 to remove and 394 not upgraded.
Need to get 10.2 MB/19.0 MB of archives.
After this operation, 51.8 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Рис. 6: Информация о виртуальной машине

Установка необходимого программного обеспечения

```
mininet@mininet-vm:~$ cd /tmp
mininet@mininet-vm:/tmp$ git clone https://github.com/ekfoury/iperf3_plotter.git
Cloning into 'iperf3_plotter' ...
remote: Enumerating objects: 74, done.
remote: Total 74 (delta 0), reused 0 (delta 0), pack-reused 74 (from 1)
Unpacking objects: 100% (74/74), 100.09 KiB | 1.11 MiB/s, done.
mininet@mininet-vm:/tmp$
```

Активация Windows

Рис. 7: Переход во временный каталог и скачивание репозитория

Установка необходимого программного обеспечения

```
mininet@mininet-vn:/tmp$ cd /tmp/iperf3_plotter
mininet@mininet-vn:/tmp/iperf3_plotter$ sudo cp plot_* /usr/bin
mininet@mininet-vn:/tmp/iperf3_plotter$ sudo cp *.sh /usr/bin
mininet@mininet-vn:/tmp/iperf3_plotter$ - Активация Windows
```

Рис. 8: Установка iperf3_plotter

```
mininet@mininet-vm:/tmp/iperf3_plotter$ sudo mn --topo=single,2 -x
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
Error starting terms: Cannot connect to display
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> _
```

Рис. 9: Задание простейшей топологии

```
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
mininet> dump
<Host h1: h1-eth0:10.0.0.1 pid=4045>
<Host h2: h2-eth0:10.0.0.2 pid=4049>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=4054>
<Controller c0: 127.0.0.1:6653 pid=4038>
mininet> _
```

Активация Windows
Чтобы активировать Wind

Рис. 10: Параметры топологии(1-я попытка)

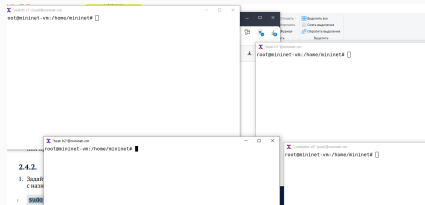
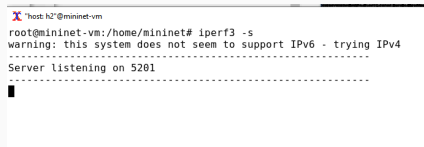


Рис. 11: Все окна терминалов есть(удачный запуск топологии)

```
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Running terms on localhost:10.0
*** Starting controller
c0
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> net
h1 h1-eth0:s1-eth1
h2 h2-eth0:s1-eth2
s1 lo: s1-eth1:h1-eth0 s1-eth2:h2-eth0
c0
mininet> links
h1-eth0<->s1-eth1 (OK OK)
h2-eth0<->s1-eth2 (OK OK)
mininet> dump
Host h1: h1-eth0:10.0.0.1 pid=853>
Host h2: h2-eth0:10.0.0.2 pid=855>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=860>
<Controller c0: 127.0.0.1:6653 pid=846>
mininet>
```

Рис. 12: Параметры топологии при 2-ой удачной попытке



```
host: h2@mininet-vm
root@mininet-vm:/home/mininet# iperf3 -s
warning: this system does not seem to support IPv6 - trying IPv4
.....
Server listening on 5201
.....
█
```

Рис. 13: Запуск сервера iPerf3 на h2

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 55500 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer      Bitrate      Retr  Cwnd
[ 7]  0.00-1.00  sec  1.23 GBytes  10.6 Gbits/sec    0   8.14 MBytes
[ 7]  1.00-2.00  sec  1.25 GBytes  10.7 Gbits/sec    1   8.14 MBytes
[ 7]  2.00-3.00  sec  1.21 GBytes  10.4 Gbits/sec    0   8.14 MBytes
[ 7]  3.00-4.00  sec  1.19 GBytes  10.2 Gbits/sec    0   8.14 MBytes
[ 7]  4.00-5.01  sec  1.21 GBytes  10.4 Gbits/sec    0   8.14 MBytes
[ 7]  5.01-6.01  sec  1.24 GBytes  10.6 Gbits/sec    0   8.14 MBytes
[ 7]  6.01-7.00  sec  1.21 GBytes  10.4 Gbits/sec    0   8.14 MBytes
[ 7]  7.00-8.00  sec  1.22 GBytes  10.5 Gbits/sec    0   8.14 MBytes
[ 7]  8.00-9.00  sec  1.21 GBytes  10.4 Gbits/sec    1   8.14 MBytes
[ 7]  9.00-10.00 sec  1.22 GBytes  10.5 Gbits/sec    0   8.14 MBytes

[ ID] Interval      Transfer      Bitrate      Retr
[ 7]  0.00-10.00  sec  12.2 GBytes  10.5 Gbits/sec    2
[ 7]  0.00-10.02  sec  12.2 GBytes  10.5 Gbits/sec

iperf Done.
root@mininet-vm:/home/mininet#
```

Рис. 14: Запуск клиента iPerf3 на h1

Интерактивные эксперименты

```
mininet> h2 iperf3 -s &
mininet> h1 iperf3 -c h2
Connecting to host 10.0.0.2, port 5201
[ 5] local 10.0.0.1 port 55594 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 5] 0.00-1.00 sec  1.17 GBytes 10.0 Gbits/sec  0    8.19 MBytes
[ 5] 1.00-2.00 sec  1.11 GBytes 9.53 Gbits/sec  0    8.19 MBytes
[ 5] 2.00-3.00 sec  1.03 GBytes 8.81 Gbits/sec  0    8.19 MBytes
[ 5] 3.00-4.00 sec  1.08 GBytes 9.26 Gbits/sec  0    8.19 MBytes
[ 5] 4.00-5.00 sec  1.15 GBytes 9.86 Gbits/sec  0    8.19 MBytes
[ 5] 5.00-6.00 sec  1.06 GBytes 9.10 Gbits/sec  0    8.19 MBytes
[ 5] 6.00-7.00 sec  1.11 GBytes 9.56 Gbits/sec  0    8.19 MBytes
[ 5] 7.00-8.00 sec  1.11 GBytes 9.54 Gbits/sec  0    8.19 MBytes
[ 5] 8.00-9.00 sec  1.11 GBytes 9.56 Gbits/sec  0    8.19 MBytes
[ 5] 9.00-10.00 sec 1.10 GBytes 9.47 Gbits/sec  0    8.19 MBytes
- - - - -
[ ID] Interval      Transfer    Bitrate      Retr
[ 5] 0.00-10.00 sec 11.0 GBytes 9.47 Gbits/sec  0          sender
[ 5] 0.00-10.02 sec 11.0 GBytes 9.45 Gbits/sec              receiver

iperf Done.
mininet> h2 killall iperf3
warning: this system does not seem to support IPv6 - trying IPv4
-----
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 55592
[ 5] local 10.0.0.2 port 5201 connected to 10.0.0.1 port 55594
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-1.00 sec  1.16 GBytes 9.56 Gbits/sec
[ 5] 1.00-2.00 sec  1.11 GBytes 9.56 Gbits/sec
[ 5] 2.00-3.00 sec  1.03 GBytes 8.84 Gbits/sec
[ 5] 3.00-4.00 sec  1.07 GBytes 9.18 Gbits/sec
[ 5] 4.00-5.00 sec  1.15 GBytes 9.88 Gbits/sec
[ 5] 5.00-6.00 sec  1.06 GBytes 9.08 Gbits/sec
[ 5] 6.00-7.00 sec  1.11 GBytes 9.50 Gbits/sec
[ 5] 7.00-8.00 sec  1.12 GBytes 9.58 Gbits/sec
[ 5] 8.00-9.00 sec  1.12 GBytes 9.66 Gbits/sec
[ 5] 9.00-10.00 sec 1.10 GBytes 9.43 Gbits/sec
[ 5] 10.00-10.02 sec 4.50 MBytes 2.30 Gbits/sec
- - - - -
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-10.02 sec 11.0 GBytes 9.45 Gbits/sec              receiver
-----
Server listening on 5201
-----
iperf3: interrupt - the server has terminated
mininet>
```

Рис. 15: Последовательность выполнения эксперимента в интерфейсе mininet

Интерактивные эксперименты

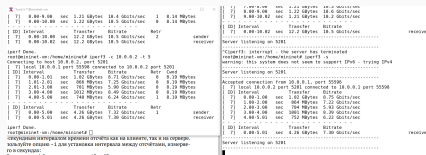


Рис. 16: Эксперимент с учетом периода времени

```
root@mininet-virtual-machine:/home/mininet# iperf3 -s -i 2
warning: this system does not seem to support IPv6 - trying IPv4
.....
Server listening on 5201
.....
```

Рис. 17: Запуск сервера iPerf3 на h2

```
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -i 2
Connecting to host 10.0.0.2, port 5201
[ 7] local 10.0.0.1 port 55602 connected to 10.0.0.2 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 7]  0.00-2.01   sec  1.99 GBytes  8.53 Gbits/sec    0   8.17 MBytes
[ 7]  2.01-4.00   sec  1.92 GBytes  8.24 Gbits/sec    0   8.17 MBytes
```

Рис. 18: Запуск клиента iPerf3 на h1

Интерактивные эксперименты

```
root@kali:~/bin# ./netperf -H 10.0.0.1 -p 5501
[ 7] 3.00-4.00 sec 1812 Mbytes 8.49 Gbits/sec 0 0.10 Mbytes
[ 7] 4.00-5.00 sec 740 Mbytes 8.24 Gbits/sec 1 0.10 Mbytes
[ 10] Interval: Transfer: Bitrate: Rate:
[ 7] 0.00-5.00 sec 4.26 Gbytes 7.22 Gbits/sec 1 sender
[ 7] 0.00-5.01 sec 4.26 Gbytes 7.18 Gbits/sec receiver

iperf Done.
root@kali:~/bin# ./netperf -H 10.0.0.1 -p 5501
Connecting to host 10.0.0.1, port 5501
[ 7] Local 10.0.0.1 port 5502 connected to 10.0.0.1 port 5501
[ 10] Interval: Transfer: Bitrate: Rate:
[ 7] 0.00-2.01 sec 1.99 Gbytes 8.23 Gbits/sec 0 0.10 Mbytes
[ 7] 2.01-4.00 sec 1.92 Gbytes 8.24 Gbits/sec 0 0.10 Mbytes
[ 7] 4.00-6.01 sec 1.84 Gbytes 8.02 Gbits/sec 0 0.10 Mbytes
[ 7] 6.01-8.00 sec 1.85 Gbytes 7.89 Gbits/sec 0 0.10 Mbytes
[ 7] 8.00-10.00 sec 1.80 Gbytes 7.75 Gbits/sec 0 0.10 Mbytes
[ 10] Interval: Transfer: Bitrate: Rate:
[ 7] 0.00-10.00 sec 9.80 Gbytes 7.73 Gbits/sec 0 sender
[ 7] 0.00-10.01 sec 9.80 Gbytes 7.72 Gbits/sec receiver

iperf Done.
root@kali:~/bin#
```

В терминале h3 запущен клиент IPerf3, задан объем данных 16 Гбайт:

```
root@kali:~/bin# ./netperf -H 10.0.0.1 -p 5501
[ 7] 0.00-5.01 sec 4.26 Gbytes 7.18 Gbits/sec receiver
Server listening on 5501
^Ciperf3: Interrupt - the server has terminated
root@kali:~/bin# ./netperf -H 10.0.0.1 -p 5501
Server listening on 5501
Accepted connection from 10.0.0.1, port 5508
[ 7] Local 10.0.0.2 port 5503 connected to 10.0.0.1 port 5502
[ 10] Interval: Transfer: Bitrate:
[ 7] 0.00-2.01 sec 1.99 Gbytes 8.23 Gbits/sec
[ 7] 2.01-4.00 sec 1.92 Gbytes 8.27 Gbits/sec
[ 7] 4.00-6.00 sec 1.83 Gbytes 7.82 Gbits/sec
[ 7] 6.00-8.00 sec 1.85 Gbytes 7.68 Gbits/sec
[ 7] 8.00-10.00 sec 1.80 Gbytes 7.77 Gbits/sec
[ 7] 10.00-10.01 sec 1.80 Mbytes 993 Mbits/sec
[ 10] Interval: Transfer: Bitrate:
[ 7] 0.00-10.01 sec 9.80 Gbytes 7.72 Gbits/sec receiver
Server listening on 5501
^Ciperf3: Interrupt - the server has terminated
root@kali:~/bin#
```

Рис. 19: Общий результат

Интерактивные эксперименты

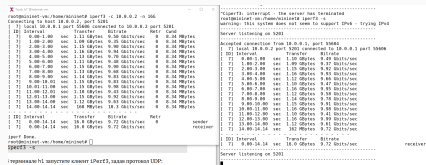
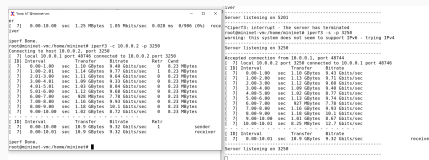


Рис. 20: Задание отправки определенного объема данных

Интерактивные эксперименты



```
Task 17 @ Windows - - - - -
net
[ 7] 0.00-10.00 sec: 1.25 Mbytes 1.85 Mbits/sec 0.028 ms 8/800 (0%) recv
peer

[peer] Done.
root@mininet-vx/home/mininet# (port3 -c 10.0.0.2 -p 3250)
Connecting to host 10.0.0.2, port 3250
[ 7] local 10.0.0.1 port 48740 connected to 10.0.0.2 port 3250
[ 1] Interval Transfer Bitrate Ratio
[ 1] 0.00-1.00 sec: 1.10 Gbytes 9.48 Gbits/sec 0 0.23 Mbytes
[ 2] 1.00-2.01 sec: 1.14 Gbytes 9.77 Gbits/sec 1 0.23 Mbytes
[ 2] 2.01-3.00 sec: 1.11 Gbytes 9.64 Gbits/sec 0 0.23 Mbytes
[ 3] 3.00-4.01 sec: 1.06 Gbytes 9.13 Gbits/sec 0 0.23 Mbytes
[ 3] 4.01-5.01 sec: 1.03 Gbytes 8.64 Gbits/sec 0 0.23 Mbytes
[ 3] 5.01-6.00 sec: 1.10 Gbytes 9.48 Gbits/sec 0 0.23 Mbytes
[ 3] 6.00-7.00 sec: 820 Mbytes 7.38 Gbits/sec 0 0.23 Mbytes
[ 3] 7.00-8.00 sec: 1.10 Gbytes 9.48 Gbits/sec 0 0.23 Mbytes
[ 3] 8.00-9.00 sec: 1.10 Gbytes 10.1 Gbits/sec 0 0.23 Mbytes
[ 3] 9.00-10.00 sec: 1.07 Gbytes 9.12 Gbits/sec 0 0.23 Mbytes
[ 12] Interval Transfer Bitrate Ratio
[ 7] 0.00-10.00 sec: 10.0 Gbytes 9.32 Gbits/sec 1 sender
[peer] Done.
root@mininet-vx/home/mininet#

[net]
Server listening on 3250
[ 1] local 10.0.0.1 port 3250 connected to 10.0.0.2 port 48740
[ 1] Interval Transfer Bitrate
[ 1] 0.00-1.00 sec: 1.10 Gbytes 9.43 Gbits/sec
[ 2] 1.00-2.00 sec: 1.13 Gbytes 9.70 Gbits/sec
[ 2] 2.00-3.00 sec: 1.13 Gbytes 9.68 Gbits/sec
[ 2] 3.00-4.00 sec: 1.09 Gbytes 9.49 Gbits/sec
[ 3] 4.00-5.00 sec: 1.02 Gbytes 8.77 Gbits/sec
[ 3] 5.00-6.00 sec: 1.13 Gbytes 9.74 Gbits/sec
[ 3] 6.00-7.00 sec: 837 Mbytes 7.76 Gbits/sec
[ 3] 7.00-8.00 sec: 1.16 Gbytes 9.92 Gbits/sec
[ 3] 8.00-9.00 sec: 1.10 Gbytes 10.1 Gbits/sec
[ 3] 9.00-10.00 sec: 1.01 Gbytes 8.67 Gbits/sec
[ 7] 10.00-10.00 sec: 8.25 Mbytes 12.7 Gbits/sec
[ 12] Interval Transfer Bitrate
[ 7] 0.00-10.00 sec: 10.0 Gbytes 9.32 Gbits/sec receiver
[peer] Done.
Server listening on 3250
```

Рис. 22: Изменение номера порта для отправки через указанный

Интерактивные эксперименты

```
[ 7] 8.88-10.68 sec 18.9 Gbytes 9.32 Gbits/sec 1 sender
[ 7] 8.88-10.61 sec 18.9 Gbytes 9.32 Gbits/sec receiver

iperf Done.
root@mininet-vn:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[ ID] Interval: Transfer: Bitrate: Retx: Cong
[ 7] 0.00-1.01 sec 1.13 Gbytes 9.62 Gbits/sec 0 0.65 Mbytes
[ 7] 1.01-2.00 sec 1.16 Gbytes 9.63 Gbits/sec 0 0.65 Mbytes
[ 7] 2.00-3.01 sec 1.11 Gbytes 9.47 Gbits/sec 0 0.65 Mbytes
[ 7] 3.01-4.00 sec 1.11 Gbytes 9.60 Gbits/sec 0 0.65 Mbytes
[ 7] 4.00-5.00 sec 1.15 Gbytes 9.92 Gbits/sec 0 0.65 Mbytes
[ 7] 5.00-6.00 sec 1.16 Gbytes 9.97 Gbits/sec 0 0.65 Mbytes
[ 7] 6.00-7.00 sec 1.14 Gbytes 9.77 Gbits/sec 0 0.65 Mbytes
[ 7] 7.00-8.00 sec 884 Mbytes 7.38 Gbits/sec 0 0.65 Mbytes
[ 7] 8.00-9.00 sec 1.13 Gbytes 9.75 Gbits/sec 0 0.65 Mbytes
[ 7] 9.00-10.68 sec 11.37 Gbytes 18.0 Gbits/sec 0 0.65 Mbytes
[ ID] Interval: Transfer: Bitrate: Retx: Cong
[ 7] 8.88-10.68 sec 11.3 Gbytes 9.52 Gbits/sec 0 sender
[ 7] 8.88-10.61 sec 11.3 Gbytes 9.52 Gbits/sec receiver

iperf Done.
root@mininet-vn:/home/mininet#
```

```
-----
*iperf3: Interrupt - the server has terminated
root@mininet-vn:/home/mininet# iperf3 -s -i
warning: this system does not seem to support IPv6 - trying IPv4
Server listening on 5201
-----
Accepted connection from 10.0.0.1, port 55614
[ ID] Interval: Transfer: Bitrate:
[ 7] 0.00-1.00 sec 1.12 Gbytes 9.59 Gbits/sec
[ 7] 1.00-2.00 sec 1.19 Gbytes 9.86 Gbits/sec
[ 7] 2.00-3.00 sec 1.18 Gbytes 9.40 Gbits/sec
[ 7] 3.00-4.00 sec 1.13 Gbytes 9.09 Gbits/sec
[ 7] 4.00-5.00 sec 1.15 Gbytes 9.66 Gbits/sec
[ 7] 5.00-6.00 sec 1.15 Gbytes 9.09 Gbits/sec
[ 7] 6.00-7.00 sec 1.15 Gbytes 9.66 Gbits/sec
[ 7] 7.00-8.00 sec 858 Mbytes 7.19 Gbits/sec
[ 7] 8.00-9.00 sec 1.16 Gbytes 9.63 Gbits/sec
[ 7] 9.00-10.68 sec 11.37 Gbytes 18.0 Gbits/sec
[ 7] 10.00-10.61 sec 1.19 Mbytes 2.10 Gbits/sec
[ ID] Interval: Transfer: Bitrate:
[ 7] 8.88-10.61 sec 11.3 Gbytes 9.52 Gbits/sec
```

Рис. 23: Задание параметра обработки данных

```
mininet@mininet-vm:~$ mkdir -p ~/work/lab_iperf3  
mininet@mininet-vm:~$ _
```

Рис. 24: Создание каталога

Интерактивные эксперименты

[illegible]

Рис. 25: Запуск сервера и запуск клиента с параметром для JSON

Интерактивные эксперименты



```
root@mininet-vm: /home/mininet# iperf3 -c 10.0.0.2 -J > /home/mininet/work/lab_i
perf3/iperf_results.json
root@mininet-vm: /home/mininet#
```

```
{
  "retransmits": 0,
  "sender": true
},
"sum_received": {
  "start": 0,
  "end": 10.011412,
  "seconds": 10.011412,
  "bytes": 12374761136,
  "bits_per_second": 9888524125.0684719,
  "sender": true
},
"cpu_utilization_percent": {
  "host_total": 49.917327002886822,
  "host_user": 0.76571061357633308,
  "host_system": 49.151626367946228,
  "remote_total": 7.4212300278197843,
  "remote_user": 0.43406274283204094,
  "remote_system": 6.9871657435333567
},
"sender_tcp_congestion": "cubic",
"receiver_tcp_congestion": "cubic"
}
```

Рис. 26: Экспорт вывода результатов

Интерактивные эксперименты

```
root@mininet-vm:~# cat /home/mininet/work/lab_iperf3/iperf_results.json
{
  "start": 0,
  "end": 10.011412,
  "seconds": 10.011412,
  "bytes": 12374761136,
  "bits_per_second": 9888524125.0684719,
  "sender": true
},
  "cpu_utilization_percent": {
    "host_total": 49.917327002886822,
    "host_user": 0.7657106135763308,
    "host_system": 49.151626367946228,
    "remote_total": 7.4212300278197043,
    "remote_user": 0.43406274283204094,
    "remote_system": 6.9871657435333567
  },
  "sender_tcp_congestion": "cubic",
  "receiver_tcp_congestion": "cubic"
}
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2 -J > /home/mininet/work/lab_iperf3/iperf_results.json
root@mininet-vm:/home/mininet# cd /home/mininet/work/lab_iperf3
root@mininet-vm:/home/mininet/work/lab_iperf3# ls -l
total 8
-rw-r--r-- 1 root root 7791 Sep 26 06:11 iperf_results.json
root@mininet-vm:/home/mininet/work/lab_iperf3#
```

Рис. 27: В каталоге присутствует iperf_results.json

```
mininet@mininet-vm:~$ xauth list $DISPLAY
mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 70c07aa5c64e2a329ca39491e7fd9bf8
mininet@mininet-vm:~$ sudo -i
root@mininet-vm:~# xauth list $DISPLAY
mininet-vm/unix:10 MIT-MAGIC-COOKIE-1 70c07aa5c64e2a329ca39491e7fd9bf8
root@mininet-vm:~# logout
mininet@mininet-vm:~$
```

Рис. 28: Права запуска X-сервера


```
mininet@mininet-vm:~$ cd ~/work/lab_iperf3
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 8
-rw-r--r-- 1 root root 7791 Sep 26 06:11 iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ sudo chown -R mininet:mininet ~/work
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 8
-rw-r--r-- 1 mininet mininet 7791 Sep 26 06:11 iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$
```

Рис. 29: Переход в каталог и коррекция прав доступа к файлу

```
mininet@mininet-vm:~/work/lab_iperf3$ plot_iperf.sh iperf3_results.json
Error: iperf3_results.json is not a file. Quitting...
mininet@mininet-vm:~/work/lab_iperf3$ plot_iperf.sh iperf3_results.json
Error: iperf3_results.json is not a file. Quitting...
mininet@mininet-vm:~/work/lab_iperf3$ ls
iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 8
-rw-r--r-- 1 mininet mininet 7791 Sep 26 06:11 iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$ plot_iperf.sh iperf_results.json
mininet@mininet-vm:~/work/lab_iperf3$
```

Рис. 30: Генерация выходных данных

Интерактивные эксперименты

```
mininet@mininet-vm:~/work/lab_iperf3$ ls -l
total 16
-rw-rw-r-- 1 mininet mininet 957 Sep 26 06:20 iperf.csv
-rw-rw-r-- 1 mininet mininet 7791 Sep 26 06:11 iperf_results.json
drwxrwxr-x 2 mininet mininet 4096 Sep 26 06:20 results
mininet@mininet-vm:~/work/lab_iperf3$ cd ~/work/lab_iperf3/results
mininet@mininet-vm:~/work/lab_iperf3/results$ ls -l
total 88
-rw-rw-r-- 1 mininet mininet 489 Sep 26 06:20 1.dat
-rw-rw-r-- 1 mininet mininet 9795 Sep 26 06:20 bytes.pdf
-rw-rw-r-- 1 mininet mininet 9618 Sep 26 06:20 cwnd.pdf
-rw-rw-r-- 1 mininet mininet 9036 Sep 26 06:20 MTU.pdf
-rw-rw-r-- 1 mininet mininet 8978 Sep 26 06:20 retransmits.pdf
-rw-rw-r-- 1 mininet mininet 8937 Sep 26 06:20 RTT.pdf
-rw-rw-r-- 1 mininet mininet 9213 Sep 26 06:20 RTT_Var.pdf
-rw-rw-r-- 1 mininet mininet 9640 Sep 26 06:20 throughput.pdf
mininet@mininet-vm:~/work/lab_iperf3/results$
```

Рис. 31: Файлы с данными и графиками успешно сформированы

Выводы

В результате выполнения данной лабораторной работы я познакомилась с инструментом для измерения пропускной способности сети в режиме реального времени — iPerf3, а также получила навыки проведения интерактивного эксперимента по измерению пропускной способности моделируемой сети в среде Mininet.